Bidirectional Transformations with Predicates*

the Dialectica reading

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Abstract

Bidirectional transformations keep two artifacts consistent, restoring consistency when either side changes. A lens is the case where one artifact is a view of the other, consistency is the graph of the read map, and the repairs write whole states rather than edits — so the consistency requirement is a fact about a lens rather than anything the lens carries. A bimorphic lens is a morphism of a dialectica category whose relation has been discarded. Restoring the relation, with values in a lineale rather than in the booleans, lets a lens carry a specification saying where it may be used. Nothing about the lens changes: there is no wrapper, no new datatype and no new way of composing, and a lens either meets a given specification or does not. The relation gives each lens a weakest precondition, and the weakest precondition of a composite is computed from those of its parts. The relation can also be read in the other direction, and then an invariant on the source determines which view edits are safe to put back, again compositionally; over a Lawvere quantale this second reading bounds how far a repair moves the source. Lenses built from total functions require *put* to be defined at every source–view pair, and the usual remedies change the type of *put* or place a monad on it, so that the laws must be restated to match. Attached to a total lens that already satisfies the laws, a predicate leaves *put* pure and the laws intact, and records instead where it is legitimate.

1 Introduction

This note is about bidirectional transformations, lenses and the Dialectica construction, and the two relations between them are not of the same kind. Lenses are a *special case* of bidirectional transformations — the asymmetric, state-based one — whereas lenses are the Dialectica construction with something *discarded*, namely the relation. The aim here is to put that back. All three are described from the beginning: the audiences for them have very few readers in common, and the lens that sits between the other two is a different object to each, so nothing below is assumed rather than said. Lenses are where we start, since they are the term common to both relations.

A lens says how a change made to a view is carried back into the source it was read from. Bidirectional transformations in general do not compose: Perdita Stevens [16] observes

*This note grew out of the 2022 Mathematics Research Community on applied category theory, where one of the working groups looked at the relationship between the Dialectica construction and lenses. That group's own account of it is Capucci, Gavranović, Malik, Rios and Weinberger [5].

that the consistency relations compose as relations but the composite has no directional maps, since recovering the intermediate artifact would need information neither side retains. Composition is available where the forward map depends on the source alone — which is the lens case, and there the composite is the one the Dialectica construction already had [6].

What the Dialectica construction has in addition is a predicate at each end and a condition saying the propagation respects it. Most lenses are not meant to be used everywhere: the update has to lie in a range, the state has to satisfy an invariant, or the write is simply undefined on some pairs. That restriction is real but it is not part of the lens. It is stated in prose beside the code, established by an argument about the particular lens, and established again about every composite the lens enters. The predicate is where it goes, and because the condition on a composite follows from the two given conditions by transitivity, the restriction is computed rather than re-established.

1.1 Bidirectional transformations

Stevens [16] makes the point that a bidirectional transformation is not primarily a transformation: it is the maintenance of consistency between two artifacts U and V. One posits a consistency relation $R \subseteq U \times V$ together with two directional transformations, each taking both artifacts and returning a repaired version of one of them. Stevens asks three things of them. *Correctness*: the repair does establish consistency. *Hippocraticness*: nothing is done when the two are already consistent. *Undoability*: reverting a change and propagating again returns the other side to what it was. She calls a transformation with all three *coherent*, and shows that very well behaved lenses are coherent in this sense. Only the first two are used below.

We take throughout the *asymmetric, state-based* case: one artifact is a view of the other, the repairs take whole states rather than edits, and R is the graph of *get*. Consistency is then functional, and therefore invisible: it is not a datum that a lens carries but a fact about it. This is worth saying because it locates the predicate of this note. A specification α is *not* a consistency relation. The two run in different directions. Consistency relates the two artifacts to each other, from U to V, and is a relation between the objects; α relates a state to a challenge to it, from U to X, within a single object, and it says how bad that challenge is rather than whether the two sides agree. Bidirectional transformations already have predicates, in the other coordinate.

They also have something closer to α than the consistency relation is, though it is not usually discussed as a predicate. In edit lenses [11] the edits form a monoid acting partially on states, and the domain of definition of that action is exactly a condition on pairs (state, edit). That is α with values in the booleans. What a lineale adds is that the condition grades rather than merely holds or fails; what §2 adds is that it composes. The other way of recording the same partiality is to change the type of *put*, as monadic lenses do [1]; the alternatives are compared in §2.

Well-behavedness in the bidirectional-transformations sense is a set of equations between round-trips — reading after writing returns what was written, and so on. Equations between maps are not conditions of the form $\alpha\colon U \times X \to L$, and the morphism condition (4) is an inequality between predicates rather than an identity between maps. So nothing in this note expresses the lens laws, and where a lens is called *conforming* below, that means it satisfies a specification α, not that it satisfies those laws.

1.2 Lenses

A lens is the abstraction of a field of a record. Given a type S of states and a type A of views, it consists of a way of reading the view out and a way of writing an updated view back in:

$$\mathit{get}\colon S \to A, \qquad \mathit{put}\colon S \times A \to S.$$

The first argument of *put* matters: writing a new value back into a record needs the old record, since everything other than the field must be carried along. Lenses compose, and a composite of a lens into a record with a lens into one of its fields is again a lens; this is what makes them useful in practice, where they are the standard way of manipulating nested immutable data.

The version used here is the *bimorphic* one, in which reading and writing may change the types:

$$\mathit{get}\colon U \to V, \qquad \mathit{put}\colon U \times Y \to X, \tag{1}$$

a lens from (U, X) to (V, Y).[1] Taking $U = X = S$ and $V = Y = A$ recovers the previous paragraph. Composition is

$$(f, F); (g, G) \;=\; \big(g \circ f,\ (u, z) \mapsto F(u, G(f(u), z))\big), \tag{2}$$

the update travelling backwards while the state travels forwards. The bidirectional transformations Stevens [16] considers, which keep no trace of the intermediate artifact, do not in general admit sequential composition using only the endpoint states. Restricting the forward map to depend on the source alone, the way a lens restricts it, gives the composition (2), and this is the Dialectica composite. Write $\mathbf{Lens}(\mathcal{C})$ for the resulting category, over any cartesian closed $\mathcal{C}$.

Lens laws — that reading after writing returns what was written, and so on — are usually imposed alongside. Nothing below needs them, and, as noted above, nothing below expresses them either; the one place they enter is the example of §2, where one of them makes the computation short.

1.3 The Dialectica construction

Gödel's Dialectica interpretation [8] assigns to a formula A a formula

$$\exists u\, \forall x\ \alpha(u, x)$$

with α quantifier-free: u ranges over *witnesses*, proposed pieces of evidence for A, and x over *counterexamples*, challenges to a proposed witness. In the lens vocabulary of §1.2 the two coordinates are the states and the updates: a witness is a state of the source, a counterexample is an update it might be asked to absorb, and α records what is being asked of it when it absorbs that one. A proof of A yields a witness; a proof of an implication yields, in addition, a way of turning a challenge to the conclusion into a challenge to the premiss.

The categorical construction takes that shape as its objects. Fix an object L of truth values — a *lineale* [12]: an order with a symmetric monoidal $\otimes$, unit e, and a right adjoint

[1] The name is Hedges' [9]. The four type parameters here are not those of a polymorphic type system: everything lives in an ordinary category with finite products. The lenses of the bidirectional-transformations literature are the further special case $U = X$ and $V = Y$, recovered by the next sentence.

$\multimap$ to it; the two-element boolean algebra is the classical case, with $\otimes = \wedge$. An object of $\mathrm{Dial}_L(\mathcal{C})$ [6, 7] is a triple (U, X, α) with $\alpha\colon U \times X \to L$, read as $\exists u \forall x\, \alpha(u, x)$. A morphism $(U, X, \alpha) \to (V, Y, \beta)$ is a pair

$$f\colon U \to V, \qquad F\colon U \times Y \to X, \tag{3}$$

subject to the condition

$$\alpha\big(u, F(u, y)\big) \;\sqsubseteq\; \beta\big(f(u), y\big), \tag{4}$$

f carrying witnesses forwards and F carrying challenges backwards. The condition is Gödel's clause for implication: whatever the challenge y to the conclusion, if u meets the challenge $F(u, y)$ manufactured from it, then $f(u)$ meets y. Reading $\alpha(u, x)$ as "u meets the challenge x" is what makes the variance come out this way, and it is the reading the relation had originally. The same data can also be read with the inequality reversed, and §3 shows that for bidirectional transformations this second reading answers a different question.

1.4 Lenses are dialectica morphisms without the relation

Displays (1) and (3) are the same data. A morphism of the Dialectica category is a lens together with a condition, and the compositions agree — (2) is the Dialectica composite. Switching the relation off makes the condition vacuous:

Lemma 1. *For $L = \mathbf{1}$, the one-element lineale, $\mathrm{Dial}_{\mathbf{1}}(\mathcal{C}) \cong \mathbf{Lens}(\mathcal{C})$, with composition agreeing on the nose.*

The containment itself has been remarked on before [10], and on its own it says little: two categories agreeing on objects and morphisms is a fact about shape until something travels along the agreement. What travels is the relation. Lenses are what remains of the Dialectica construction once it is discarded, and the rest of the note is about what it does when it is put back. By Lemma 1 we may speak of a lens and a relation-free dialectica object interchangeably.

2 Where a specification should live

Suppose a lens is not meant to be used everywhere. The update may have to lie in a range, the state may have to satisfy an invariant, or *put* may simply be undefined on some pairs. The condition has to be recorded somewhere, and there are only a few places to put it. It can go into the types, at the cost of needing dependent ones. It can go into the code, as a wrapper that returns an option or an error, at the cost of changing what a lens is and having to say what the laws now mean. Or it can be left where it belongs, as a statement about a lens that is either true or false of it.

The third is what a dialectica object provides. A specification for a lens with states U and updates X is exactly a map $\alpha\colon U \times X \to L$: how bad it is to update state u by x. Condition (4) then says that the lens respects the specifications at its two ends. So a lens does not become a different thing when it acquires a specification. There is no new datatype and no new composition operator; the composite of two lenses is the composite it always was, and the only question is which specifications it respects. This needs *put* to be total. A *put* that

is genuinely undefined somewhere has first to be extended, by a default value for instance, and the predicate then separates the legitimate uses of that extension from the rest. The extension need not satisfy the lens laws everywhere: the predicate can exclude the uses where a law fails, but it cannot make the law hold there.

What makes this workable rather than merely tidy is that the question has a best answer, and that the best answers compose. Fix a lens $\ell = (f, F)$ — the pair of (1), abbreviated to a single letter for this section — from (U, X) to (V, Y). From here on the base is **Set** and L is a complete lineale, so the meets and joins below are computed elementwise; over a more general base they would need L-valued predicates to have adjoints to reindexing, as in a hyperdoctrine. Given a specification β on the target, ask which specification on the source ℓ could possibly respect. A state u can be challenged by x only through some target challenge y with $F(u, y) = x$, so the demand β makes on (u, x) is whatever all such y demand together:

$$\ell^{\#}\beta(u, x) \;=\; \bigwedge\{\beta(f(u), y) : F(u, y) = x\}.$$

This is the weakest specification on the source that ℓ carries into β — a precondition, computed from a postcondition and a program. If no y has $F(u, y) = x$ the meet is over the empty set and equals $\top$: an update the lens can never produce at u is unconstrained. Weakest and greatest describe the same predicate: specifications are ordered by implication, so a weaker condition is a larger element of the order, and $\ell^{\#}\beta$ is the largest α for which ℓ is a morphism into β. In the other direction, a specification α on the source is pushed forward to the strongest one the lens can guarantee downstream,

$$\ell_!\alpha(v, y) \;=\; \bigvee\{\alpha(u, F(u, y)) : f(u) = v\},$$

and the two are adjoint.

Theorem 2. *$\ell_! \dashv \ell^{\#}$, and the lens ℓ underlies a morphism $(U, X, \alpha) \to (V, Y, \beta)$ of $\mathrm{Dial}_L(\mathbf{Set})$ if and only if $\alpha \sqsubseteq \ell^{\#}\beta$, equivalently $\ell_!\alpha \sqsubseteq \beta$. Moreover $(\ell'\ell)^{\#} = \ell^{\#}\ell'^{\#}$.*[2]

The specification is a precondition, and preconditions compose. This has three readings for a programmer. First, the condition is a Hoare triple for a bidirectional program: $\ell^{\#}\beta$ is the weakest precondition, and asking whether a lens meets a specification is asking whether its actual precondition is implied by the one you were willing to assume. Second, because the fibres are posets, there is nothing to choose — a lens meets a specification or it does not, and no data is being carried around to record how. Third, $(\ell'\ell)^{\#} = \ell^{\#}\ell'^{\#}$ says that the obligation attaching to a composite is obtained from the obligations of the parts, by composing them in the reverse order. Verification travels with composition instead of starting again at every stage, which is the property a library needs if specifications are to be checked at all.

Example 3. The composers lists of Bohannon, Foster, Pierce, Pilkiewicz and Schmitt [3] give a lens where everything can be computed. A source is a list of records (*name*, *dates*, *nationality*) with distinct names, and the view forgets the dates. Take $U = X$ to be the source lists and

[2] The same thing in the language of fibrations, for readers who want it: the forgetful functor $\mathrm{Dial}_L(\mathbf{Set}) \to \mathbf{Lens}(\mathbf{Set})$ is a faithful bifibration with posetal fibres. The fibre over an object (U, X) is the poset of specifications $U \times X \to L$ under the pointwise order, a lens induces the two adjoint predicate transformers above between the fibres over its ends, and faithfulness says that a lens respects a specification in at most one way.

$V = Y$ the view lists. The map *get* deletes the dates, and $put(u, y)$ is the list y with each record given the dates u has for the same name, or the placeholder ? when u has no such name. This lens satisfies $get(put(u, y)) = y$.

As a specification on the view take the policy that no composer is deleted: over $L = \mathbf{2}$, let $\beta(v, y) = \top$ when every name in v occurs in y, and $\bot$ otherwise. Since $get(put(u, y)) = y$, the only view list that can produce a source list x is $get(x)$, so the meet defining $\ell^{\#}\beta(u, x)$ has at most one term. It has one exactly when $x = put(u, get(x))$, that is when x gives every name the dates u has for it and gives ? to each new name; then $\ell^{\#}\beta(u, x) = \beta(get(u), get(x))$, which holds when x keeps every name of u. Every other x gets $\top$. So the precondition is the view policy read through *get*, on the updates the lens can produce, and it says nothing about updates that change dates, since the lens never produces one.

Now compose with a second lens ℓ' from view lists to lists of names, whose *put* restores each nationality by name and writes *unknown* for a new name, and impose on the names the same policy β' that none is deleted. The same computation gives $\ell'^{\#}\beta'(v, y)$ as the condition that y keeps every name of v, on the y that restore nationalities from v, and $\top$ elsewhere. Applying $\ell^{\#}$ to this gives, at (u, x), the condition that x keeps every name of u, on the x that restore both dates and nationalities from u, and $\top$ elsewhere. Computing $(\ell'\ell)^{\#}\beta'$ directly from the composite lens, whose *put* restores dates and nationalities by name, gives the same predicate, as Theorem 2 says it must.

In the example the meet has at most one term, and this holds for any lens with $get(put(u, y)) = y$: for such a lens the weakest precondition is the target specification read through *get*, on the image of $put(u, -)$. The meet does real work for lenses without that law, such as one whose *put* normalises its input — sorting the list by name, say — so that several view lists produce the same source list and the obligation on it is what all of them demand together.

Over a Lawvere quantale the precondition is a charge. Take $L = ([0, \infty], \geq, +, 0)$ [13]. The order $\sqsubseteq$ is the numerical $\geq$, so the meet of a set of values is its supremum and the meet of the empty set is 0. Condition (4) reads $\alpha(u, F(u, y)) \geq \beta(f(u), y)$: whatever a view update is charged, the source update it produces is charged at least as much. The weakest precondition charges each source update the most that any view update producing it is charged, and charges nothing for an update the lens never produces; this is the least charge on the source that the lens admits, the largest predicate in $\sqsubseteq$ being numerically the smallest. Theorem 2 computes the charge on a composite from the charges on its parts. The charge bounds the source side from below. The bound in the other direction, on how far a repair moves the source, is the second reading of §3.

3 Two readings

The morphism condition (4) carries satisfaction forwards: if a state meets the challenge that F manufactures from y, then its image meets y. Gödel's clause can also be read the other way round, carrying satisfaction backwards from the view to the source, and for bidirectional transformations the two readings answer different questions. The first asks which source updates a policy on the view demands, as in Example 3. The second asks which view edits are safe to put back, given an invariant on the source, which is the question of the view-update problem. This section sets out the second reading, gives it a best specification of its own, computed forwards along the lens, and shows that this specification composes as well.

The second reading asks of a lens (f, F) from (U, X, α) to (V, Y, β) that

$$\beta\big(f(u), y\big) \;\sqsubseteq\; \alpha\big(u, F(u, y)\big) \tag{5}$$

for all u and y: whenever the view update y is legitimate at $f(u)$, the source update that F produces from it is legitimate at u. Such lenses compose. Given a second lens (g, G) from (V, Y, β) to (W, Z, γ) satisfying (5), the condition for (g, G) at $f(u)$ and z, followed by the condition for (f, F) at u and $G(f(u), z)$, gives

$$\gamma\big(gf(u), z\big) \sqsubseteq \beta\big(f(u), G(f(u), z)\big) \sqsubseteq \alpha\big(u, F(u, G(f(u), z))\big),$$

which is (5) for the composite (2). Write $\mathrm{Dial}_L^{co}(\mathbf{Set})$ for the resulting category. Its objects are those of $\mathrm{Dial}_L(\mathbf{Set})$, its morphisms are lenses as before, and only the order in the condition has changed. Over the two-element lineale nothing new appears: negating the relations, $(U, X, \alpha) \mapsto (U, X, \neg\alpha)$, turns (5) into (4) and leaves the lenses untouched, so the second reading of a boolean specification is the first reading of its negation. Over a Lawvere quantale the two readings say different things, as the end of this section shows.

For a lens $\ell = (f, F)$ from (U, X) to (V, Y), over **Set** and a complete lineale L as in §2, put

$$\ell^\flat\alpha(v, y) \;=\; \bigwedge\{\, \alpha(u, F(u, y)) : f(u) = v \,\}, \qquad \ell_\flat\beta(u, x) \;=\; \bigvee\{\, \beta(f(u), y) : F(u, y) = x \,\}.$$

The first takes a specification on the source to one on the view; at a view v outside the image of f the meet is empty and equals $\top$, so a view that no source produces is unconstrained.

Theorem 4. *The map $\ell_\flat$ is left adjoint to $\ell^\flat$, and the lens ℓ underlies a morphism $(U, X, \alpha) \to (V, Y, \beta)$ of $\mathrm{Dial}_L^{co}(\mathbf{Set})$ if and only if $\beta \sqsubseteq \ell^\flat\alpha$, equivalently $\ell_\flat\beta \sqsubseteq \alpha$. Moreover $(\ell'\ell)^\flat = \ell'^\flat\,\ell^\flat$.*

Proof. Unfolding the meet, $\beta \sqsubseteq \ell^\flat\alpha$ says that $\beta(v, y) \sqsubseteq \alpha(u, F(u, y))$ whenever $f(u) = v$, which is (5). Unfolding the join, $\ell_\flat\beta \sqsubseteq \alpha$ says that $\beta(f(u), y) \sqsubseteq \alpha(u, x)$ whenever $F(u, y) = x$, which is (5) again. The two conditions are therefore equivalent for all α and β, which is the adjunction. For the composite of ℓ with $\ell' = (g, G)$,

$$\ell'^\flat\ell^\flat\alpha(w, z) = \bigwedge_{g(v)=w} \bigwedge_{f(u)=v} \alpha\big(u, F(u, G(v, z))\big) = \bigwedge_{gf(u)=w} \alpha\big(u, F(u, G(f(u), z))\big) = (\ell'\ell)^\flat\alpha(w, z),$$

since the backward map of the composite is $(u, z) \mapsto F(u, G(f(u), z))$. □

The four operations range over the same two kinds of fibre. The weakest precondition $\ell^\#$ and the new $\ell_\flat$ take the meet and the join over the fibres of $F(u, -)$; $\ell_!$ and $\ell^\flat$ take the join and the meet over the fibres of f. The first reading computes its best specification backwards along the lens, so that its composition law reverses the order of the parts, while $\ell^\flat$ is computed forwards and composes in the order of the lenses themselves.

Example 5. Return to the composers lens ℓ of Example 3, and take on the source the invariant that an update creates no new placeholder: $\alpha(u, x) = \top$ when every record of x whose dates are ? belongs to a name that already had ? in u, and $\bot$ otherwise. The update $put(u, y)$ copies the dates of every name that u has and writes ? for every other name, so $\alpha(u, put(u, y)) = \top$ exactly when every name in y occurs in u. The names in u are the

names in $get(u)$, so the meet defining $\ell^\flat\alpha(v,y)$ takes the same value at every u in the fibre: $\ell^\flat\alpha(v,y) = \top$ when every name in y occurs in v. The safe view edits delete and reorder composers but add none, which is the mirror image of the policy of Example 3. Composing with the lens ℓ' that forgets nationalities, the same computation gives $\ell'^\flat\ell^\flat\alpha$ on lists of names as the condition that the edited list adds no name, and computing $(\ell'\ell)^\flat\alpha$ directly from the composite lens gives the same predicate, as Theorem 4 says it must.

Over the Lawvere quantale $L = ([0,\infty], \geq, +, 0)$ the second reading gives the bound that the first does not. The order $\sqsubseteq$ is the numerical $\geq$, so (5) reads $\beta(f(u), y) \geq \alpha(u, F(u,y))$: the source update a repair produces is charged at most what the view update was charged. When the states carry a distance d and $\alpha(u,x) = d(u,x)$ measures how far an update moves the state, a morphism of $\mathrm{Dial}_L^{co}(\mathbf{Set})$ is a lens whose repairs move the source no further than the view edit is charged, and when the charge is itself a distance on views the repairs are nonexpansive. This bounds the change made by the repair; it does not say that the repair is the smallest change among the sources with the requested view, which would need a further hypothesis. The meet in $\sqsubseteq$ is the numerical supremum, so $\ell^\flat\alpha(v,y)$ is the largest distance by which putting y back moves any source with view v, and is 0 at a view that no source produces. Theorem 4 computes this bound on a composite from the bounds on its parts.

4 What a predicate buys

The fibrational and profunctorial accounts of optics — lenses as morphisms of fibrations [15], optics as Tambara modules [14, 4], and the treatment of lenses, optics and dialectica categories as instances of one twisted tower of fibrations [5] — are considerably more general than the setting used here, and they answer questions this note does not ask. The setting here is the narrow one on purpose: an ordinary category with finite products, a fixed pair of types at each end, and a predicate on them. Everything above is a statement about lenses of that shape, which is the shape of the lenses people write.

A specification has so far been attached to a lens from outside: given ℓ and β, one asks whether ℓ meets it. A library wants the other thing too — to speak of the lenses meeting a given specification as a type, so that a conforming lens can be stored, passed to a function, or returned by one. That asks for the collection of them to be an object of the same category. Bodil Biering [2] gets close, using Hyland's comonad on the update type: with updates in $X + 1$ the backward map may decline, returning no update rather than one it cannot justify, and the option of declining is what makes a function space available at all. What she obtains is a weak exponential, a retraction rather than an adjunction, cartesian closure arriving only after the idempotent is split. The gap is the candidates that meet no condition, and closing it asks for an operation on the *first* coordinate: cutting the witnesses down to those satisfying their own specification, which is what subset types are for in the syntactic setting. That operation and the function space it yields are taken up elsewhere.

Theorem 2 is what the predicate buys. A specification can be carried without changing the lens, and the obligation on a composite is computed from the obligations on its parts, so a precondition need not be propagated by hand and need not be re-established at every stage. The same formula computes charges when the lineale is a Lawvere quantale, and the composers example shows the computation on a lens from the bidirectional-transformations literature. Theorem 4 does the same for the second reading: an invariant on the source

determines which view edits are safe to put back, the condition on a composite is computed from the conditions on its parts, and over a Lawvere quantale the same computation bounds how far a repair moves the source.

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